Muhammed Salih DEDE

CS 4th Year Student | Computer Engineering

GitHub | LinkedIn | Hugging Face | Kaggle | Medium

Contact Me

0535 884 81 82
dedem@mef.edu.tr
Avcılar / İstanbul

About Me

> As an AI Engineer with a focus on architecting production-grade autonomous agents, I specialize in transforming complex, real-world workflows into reliable, scalable multi-agent systems. I leverage deep expertise in LangGraph and LangChain to build sophisticated orchestration logic, ranging from creative content pipelines to personality-driven conversational models. By integrating advanced LLM techniques with robust tool-use frameworks, I am committed to delivering high-performance, end-to-end AI solutions that drive measurable impact for users.

Experience

AI Engineer at Codeventure — 2026 – present
Architect and deploy production-grade autonomous agents using LangGraph and LangChain. Optimize agentic workflows for reliability and performance, integrating n8n for scalable automation across complex business processes.

AI Engineering Intern at Codeventure — 2025 – 2026
Assisted in the development of LLM-based agentic systems. Implemented orchestration logic and automated workflows using LangGraph and LangChain to streamline internal business operations.

AI Engineering Intern at Microsoft — 2025
Contributed to the development of neural network architectures and machine learning models. Focused on optimizing model performance and data processing pipelines for AI-driven applications.

Education

MEF University — 2021 – present
Bachelor of Science, Computer Engineering — GPA: 3.58 / 4.0
4th year student

Florya Final Anadolu High School — 2018 – 2019
High School Diploma
Graduate with a high degree. Top student of the school.

Skills and Language

Python	■■■■■	TypeScript	■■■□□	SQL / PostgreSQL	■■■□□
Docker	■■■□□	Git / GitHub	■■■■■	GraphQL	■■■■■
LangChain	■■■■■	LangGraph	■■■■■	n8n	■■■■■
PyTorch	■■■□□	Hugging Face Transformers	■■■□□	LoRA / PEFT	■■■□□
ChromaDB	■■■□□	Playwright	■■■□□	LangSmith	■■■■■
Pinecone	■■■■■	Kubernetes	■■□□□	Redis	■■□□□

Turkish — native **English** — professional

Projects

Hacivat & Karagöz — AI Shadow Play — 2026
Built an autonomous AI shadow play pipeline using FastAPI and Gemini 2.0. Implemented real-time voice synthesis and synchronized animation engines to deliver a fully automated, multi-modal performance system.
Tech Stack : Python - FastAPI - PIXI.js - XTTS v2 - Gemini 2.0 Flash - OpenRouter - PyTorch - TTS

Fincan Kahvem Hatrına — Turkish RAP AI — 2026 Demo Github Repo
Orchestrated a multi-agent system using LangGraph to automate creative workflows including lyrics critique and refinement. Utilized LoRA for fine-tuning and synthetic data generation to improve model performance and output quality.
Tech Stack : Python - LoRA - LangGraph - Mistral - HuggingFace - Synthetic Data Generation - PyTorch - RunPod

Kumru - Turkish NLP Character Chatbots 2026

Developed personality-driven chatbots using fine-tuned LLMs and LoRA/PEFT techniques. Managed the end-to-end pipeline from data scraping and N-gram analysis to deployment and inference optimization.

Tech Stack : Python - Hugging Face Transformers - LoRA - PEFT - NLTK - PyTorch

LinkedIn Automation System with LLM Integration 2025

Architected an intelligent sales automation system featuring human-in-the-loop (HITL) workflows. Integrated LangChain with Playwright to automate personalized outreach while maintaining strict control over agent-generated content.

Tech Stack : Python - Playwright - LangChain - SQL - Telegram API

Hulki (AI Agent) 2025

Built 'Hulki', an AI agent leveraging LangGraph for stateful task orchestration and ChromaDB for RAG-based knowledge retrieval. Implemented real-time attention tracking and multi-modal analysis to provide personalized user assistance.

Tech Stack : Python - Vision - NLP - React - ChromaDB - LangChain - LangGraph

Certificates

Microsoft AI Innovators Internship Program	2025-08-11
OpenAI Assistant: Create a Code to UML-Diagram Generator	2025-05-12
GitHub Copilot for Beginners: Write Software With AI	2025-05-11